

E-Commerce Intelligence System (SQL-Based Business Analytics Engine)

Project Overview

This project analyzes 100,000+ e-commerce transactions to build a complete business intelligence system using SQL. The goal was to transform raw transactional data into actionable insights that reveal growth drivers, profitability risks, customer value patterns, and operational inefficiencies.

Business Objectives

- Analyze revenue growth trends and seasonality patterns
- Measure Average Order Value (AOV) behavior over time
- Evaluate margin impact caused by freight costs
- Segment customers into high-value and low-value groups
- Identify repeat customer contribution to total revenue
- Detect product revenue concentration risks
- Build early churn risk indicators

Key KPIs Developed in SQL

- Total Revenue & Monthly Growth Rate
- Average Order Value (AOV)
- Customer Lifetime Value (CLV)
- Repeat Purchase Rate
- Revenue per Customer
- Freight Cost Ratio
- Product Revenue Contribution Percentage
- Churn Risk Segmentation Metrics

Technical Implementation

The system was built entirely using SQL with advanced concepts such as multi-table joins, Common Table Expressions (CTEs), window functions, aggregation logic, ranking functions,

case-based segmentation, and retention analysis. The project simulates a real-world analytics environment where SQL is used as a complete business intelligence engine without relying on external tools.

Key Business Insights

- A small percentage of customers contribute the majority of revenue (Pareto effect)
- Freight costs significantly reduce net margins in low-value orders
- Revenue growth does not always align with AOV growth
- Product dependency risk exists when few products dominate sales
- Repeat customers form the core revenue engine of the business

Conclusion

This project demonstrates how SQL alone can be used to design and implement a full-scale E-Commerce Intelligence System. It reflects real-world analytics workflows used by business intelligence and data teams to monitor performance, detect risks, and support strategic decision-making.